

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Debugging

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 25%

Total Marks: 20

Code of Conduct

I Hariharan G(192424156) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: HARIHARAN G

Debugging

Criteria	Anomaly Detection (4 Marks)	Cleaning Methods (4 Marks)	Data Transformation (4 Marks)	Analytical Comparison (4 Marks)	Governance Discussion (4 Marks)
Marks					

Total Marks Awarded:

Questions:

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (4 Marks)

2. Design a cleaning strategy for each issue. (4 Marks)
3. Perform data standardization and transformation. (4 Marks)
4. Analyze how cleaning affects total revenue calculations. (4 Marks)
5. Generate a report discussing business risks if the data were not cleaned (4 Marks)

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues

- Missing patient IDs

- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (4 Marks)
2. Develop rules for handling missing and invalid values. (4 Marks)
3. Create a process to standardize diagnosis names. (4 Marks)
4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (4 Marks)
5. Recommend data governance measures to prevent future errors. (4 Marks)

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved

L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (4 Marks)
2. Propose cleaning techniques for missing and inconsistent data. (4 Marks)
3. Detect and justify treatment of outliers. (4 Marks)
4. Analyze the effect of cleaning on loan approval rates. (4 Marks)
5. Discuss ethical implications of incorrect data cleaning decisions. (4 Marks)

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers

- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (4 Marks)
2. Recommend cleaning techniques for time-series data. (4 Marks)
3. Design rules to identify faulty sensor readings. (4 Marks)
4. Evaluate how cleaning affects congestion analysis. (4 Marks)
5. Suggest automated monitoring techniques for future data collection. (4 Marks)

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (4 Marks)
2. Design methods to standardize currencies, products, and dates. (4 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (4 Marks)
4. Compare regional sales rankings before and after cleaning. (4 Marks)
5. Prepare an executive report explaining how data quality influences strategic business decisions. (4 Marks)

Signature of the Course Faculty

Absolutely. Below is ready-to-copy assessment report content for all 5 questions. You can paste this into Word and insert your Colab output screenshots in the marked spaces.

The structure follows your assessment: identify issues → cleaning strategy → transformation → analysis → business/governance discussion.

ASSESSMENT TOOL 3 – DEBUGGING

COURSE NAME

Query Processing for Data Science

COURSE CODE

DSA0503

ASSESSMENT TOOL

Assessment Tool 3 – Debugging

TITLE

Data Cleaning Challenge

COURSE OUTCOME

CO2: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

QUESTION 1: E-COMMERCE CUSTOMER TRANSACTIONS DATASET

1. Aim

To identify and clean data-quality issues in an e-commerce customer transaction dataset and analyze how data cleaning affects total revenue calculations.

2. Dataset Description

The dataset contains customer transaction information including:

- Customer ID
- Customer Name
- Gender
- Age
- Product Category
- Quantity

- Unit Price
- Purchase Date
- City

The dataset contains several quality problems such as missing Customer IDs, duplicate transactions, negative quantities, different currencies, invalid ages, inconsistent city names, and incorrect dates. These issues are specified in the given assessment dataset.

3. Data Quality Issues Identified

Issue	Example	Type
Missing Customer ID	NULL	Missing value
Duplicate transaction	C105	Duplicate
Invalid age	0, 150	Outlier/invalid
Negative quantity	-2	Invalid value
Multiple currencies	INR, USD, EUR	Formatting/standardization
City inconsistency	Chennai, chennai, CHENNAI	Inconsistency
Invalid date	2025-13-01	Invalid format/value

4. Cleaning Method

The following cleaning operations were performed:

1. Duplicate transactions were removed.
2. Missing Customer ID was assigned a temporary documented identifier.
3. Ages outside the range 1–120 were converted to missing values.
4. Negative quantities were treated as invalid.
5. City names were standardized using string formatting.
6. Price and currency were separated from the UnitPrice column.
7. Currency values were converted to INR for revenue comparison.
8. Purchase dates were converted into a standard date format.
9. Revenue was calculated using quantity multiplied by standardized unit price.

5. Code Explanation

The Python program uses Pandas and NumPy for data cleaning.

`drop_duplicates()` removes duplicate transactions.

`fillna()` handles missing Customer IDs.

`where()` identifies invalid age and quantity values.

`str.title()` standardizes city names.

Regular expressions are used to separate the numeric price and currency.

The `pd.to_datetime()` function converts purchase dates into a standard date format.

Finally, revenue is calculated using:

$$\text{Revenue} = \text{Quantity} \times \text{Unit Price}$$

6. Output

```
===== DATA QUALITY CHECK =====
Missing Values:
CustomerID      1
CustomerName    0
Gender          0
Age            1
ProductCategory 0
Quantity        0
UnitPrice       0
PurchaseDate    0
City            0
dtype: int64

Duplicate Rows:
1

Raw Revenue:
₹ 250200

===== CLEANED DATASET =====
```

	CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City	Price	Currency	UnitPriceINR	RevenueINR
0	C101	Raj Kumar	Male	28.0	Electronics	2.0	15000 INR	2025-01-10	Chennai	15000	INR	15000	30000.0
1	C102	Priya S	Female	35.0	Fashion	3.0	1200 INR	2025-01-11	Chennai	1200	INR	1200	3600.0
2	C103	Arun V	Male	NaN	Electronics	1.0	300 USD	2025-01-12	Chennai	300	USD	24900	24900.0
3	C104	Kavya R	Female	NaN	Grocery	NaN	800 INR	2025-01-15	Coimbatore	800	INR	800	NaN
4	C105	Rahul P	Male	42.0	Fashion	5.0	2500 INR	2025-01-18	Madurai	2500	INR	2500	12500.0
6	MISSING_CID_001	Sneha M	Female	30.0	Electronics	2.0	450 USD	2025-02-05	Trichy	450	USD	37350	74700.0
7	C107	Deepak N	Male	NaN	Grocery	3.0	600 INR	2025-02-07	Tiruchirappalli	600	INR	600	1800.0
8	C108	Meena K	Female	27.0	Fashion	1.0	1800 INR	NaN	Salem	1800	INR	1800	1800.0
9	C109	John D	Male	31.0	Electronics	4.0	250 EUR	2025-02-15	Chennai	250	EUR	22500	90000.0

```
Number of rows after cleaning: 9

===== REVENUE ANALYSIS =====
Total Revenue After Cleaning:
₹ 239300.0

Revenue by Product Category:
RevenueINR
ProductCategory
Electronics    219600.0
Fashion        17900.0
Grocery         1800.0

dtype: float64
```

```
===== REVENUE ANALYSIS =====
Total Revenue After Cleaning:
₹ 239300.0

Revenue by Product Category:
RevenueINR
ProductCategory
Electronics    219600.0
Fashion        17900.0
Grocery         1800.0

dtype: float64
```

7. Result

The e-commerce dataset was successfully cleaned by removing duplicates, handling invalid values, standardizing cities and dates, and converting currencies. The cleaned data provides a more reliable basis for calculating customer and product revenue.

QUESTION 2: HOSPITAL PATIENT RECORDS DATASET

1. Aim

To detect inconsistencies in hospital patient records, clean invalid values, standardize diagnosis names, and analyze the effect of cleaning on treatment cost and hospital stay duration.

2. Dataset Description

The dataset contains:

- Patient ID
- Patient Name
- Age
- Gender
- Admission Date
- Discharge Date
- Diagnosis
- Treatment Cost
- Doctor ID

The assessment identifies missing patient IDs, duplicate records, invalid discharge dates, missing treatment costs, inconsistent diagnosis names, multiple date formats, and unrealistic ages.

3. Data Quality Issues

Issue	Example	Type
Missing Patient ID	NULL	Missing
Duplicate record	P1005	Duplicate
Invalid age	125	Invalid/outlier

Issue	Example	Type
Invalid date sequence	Discharge before admission	Logical error
Missing cost	NULL	Missing
Diagnosis variation	Diabetes/diabetes/Diabtes	Inconsistency
Date formatting	Different date formats	Formatting

4. Cleaning Method

The following operations were performed:

1. Duplicate records were removed.
2. Missing Patient ID was replaced with a documented identifier.
3. Ages above 120 were treated as invalid.
4. Admission and discharge dates were converted to standard datetime format.
5. Diagnosis names were converted to lowercase.
6. The spelling error Diabtes was corrected to diabetes.
7. Hospital stay duration was calculated.
8. Negative hospital stays were treated as invalid.

5. Code Explanation

`drop_duplicates()` removes repeated patient records.

`fillna()` handles the missing Patient ID.

`between()` checks whether patient ages are within a valid range.

`pd.to_datetime()` standardizes admission and discharge dates.

Diagnosis values are converted to lowercase to eliminate case-based inconsistencies.

The spelling variation Diabtes is mapped to diabetes.

Hospital stay is calculated as:

Hospital Stay = Discharge Date – Admission Date

6.output

```
Name      0
Age       1
Gender    0
AdmissionDate 0
DischargeDate 0
Diagnosis 0
TreatmentCost 2
DoctorID  0
dtype: int64

Duplicate Rows:
1

***** BEFORE CLEANING *****
Average Treatment Cost: 16937.5
Average Hospital Stay: 4.5 days

***** CLEANED DATASET *****
```

	PatientID	Name	Age	Gender	AdmissionDate	DischargedDate	Diagnosis	TreatmentCost	DoctorID	StayDays
0	P1001	Ravi Kumar	45.0	M	2025-01-10	2025-01-15	diabetes	25000.0	D101	5.0
1	P1002	Priya S	38.0	F	2025-01-12	2025-01-11	hypertension	18000.0	D102	NaN
2	P1003	Arun V	NaN	M	2025-01-13	2025-01-20	diabetes	22000.0	D101	7.0
3	MISSING_PID_001	Kavya R	29.0	F	2025-01-15	2025-01-17	diabetes	15000.0	D103	2.0
4	P1005	Rahul P	51.0	M	2025-01-18	2025-01-25	asthma	NaN	D104	7.0
6	P1006	Sneha M	34.0	F	2025-01-15	2025-01-18	hypertension	12000.0	D102	3.0
7	P1007	Deepak N	NaN	M	2025-02-01	2025-02-05	asthma	10000.0	D104	4.0
8	P1008	Meena K	26.0	F	2025-02-10	2025-02-15	hypertension	13500.0	D102	5.0
9	P1009	John D	40.0	M	2025-02-12	2025-02-18	diabetes	20000.0	D101	6.0

```
Rows after cleaning: 9
```

7. Result

The hospital dataset was successfully cleaned. Duplicate records and invalid ages were handled, diagnosis names were standardized, and invalid hospital stays were identified. The cleaned dataset provides more reliable treatment-cost and hospitalization analysis.

QUESTION 3: BANKING LOAN APPROVAL DATASET

1. Aim

To identify data-quality problems in banking loan records, clean inconsistent and invalid values, detect extreme outliers, and evaluate the effect of cleaning on loan approval rates.

2. Dataset Description

The dataset contains:

- Customer ID
- Annual Income
- Credit Score
- Loan Amount
- Employment Status
- Age

- Existing Debt
- Loan Status

The assessment identifies missing credit scores, different income units, negative loan amounts, duplicate customer IDs, invalid ages, inconsistent employment status, and extreme income values.

3. Data Quality Issues

Issue	Example	Type
Missing credit score	NULL	Missing
Different income representation	900K	Formatting
Negative loan amount	-200000	Invalid
Duplicate ID	L008	Duplicate
Invalid age	16	Invalid
Employment variation	FT, Full Time, fulltime	Inconsistency
Extreme income	75000000	Outlier

4. Cleaning Method

1. Duplicate records were removed.
2. 900K was converted to 900000.
3. Missing credit score was handled using the median.
4. Missing annual income was handled using the median.
5. Employment status values were standardized.
6. Negative loan amounts were removed.
7. Applicants below 18 years were removed.
8. The extreme income value was treated as an outlier for this demonstration.
9. Loan approval rate was calculated before and after cleaning.

5. Code Explanation

The code first loads the banking CSV using Pandas.

`drop_duplicates()` removes duplicate loan records.

The `replace()` function converts 900K into a numerical income value.

Missing credit scores are filled using the median credit score.

Employment status values such as FT and fulltime are converted to the standard value Full Time.

Invalid loan amounts and ages are removed.

The approval rate is calculated as:

$\text{Approval Rate} = \text{Approved Applications} / \text{Total Valid Applications} \times 100$

6. Output

```
***** DATA QUALITY CHECK *****
Missing Values:
CustomerID      0
AnnualIncome    1
CreditScore     1
LoanAmount      0
EmploymentStatus 0
Age             0
ExistingDebt    0
LoanStatus      0
dtype: int64

Duplicate Rows:
1

Approval Rate Before Cleaning:
80.0 %

***** CLEANED DATASET *****
CustomerID AnnualIncome CreditScore LoanAmount EmploymentStatus Age ExistingDebt LoanStatus
0      L001      750000.0        720.0      300000          Full Time    35         50000    Approved
1      L002      900000.0        680.0      400000          Full Time    42        100000    Approved
2      L003     1200000.0        700.0      600000          Full Time    28        200000    Approved
5      L006      450000.0        350.0      250000          Unemployed    40        300000    Rejected
7      L008      650000.0        690.0      350000          Full Time    33         70000    Approved
9      L010      825000.0        710.0      300000          Full Time    37         80000    Approved

Rows after cleaning: 6

***** APPROVAL ANALYSIS *****
Approval Rate Before Cleaning: 80.0 %
Approval Rate After Cleaning: 83.33 %

Loan Status Distribution:
count
```

7. Result

The banking dataset was successfully cleaned by handling missing values, duplicates, inconsistent employment status, invalid loan amounts, invalid ages, and extreme income values. The resulting approval rate provides a more meaningful representation of valid loan applications.

8. Ethical Consideration

Data cleaning in loan approval systems must be performed carefully because incorrect removal or modification of applicant information can directly affect financial decisions. Cleaning rules should therefore be transparent, auditable, and applied consistently.

QUESTION 4: SMART CITY TRAFFIC MONITORING DATASET

1. Aim

To assess and clean traffic sensor data, identify faulty sensor readings, standardize time-series information, and evaluate how cleaning affects traffic and congestion analysis.

2. Dataset Description

The dataset contains:

- Sensor ID
- Timestamp
- Vehicle Count
- Average Speed
- Traffic Signal Status
- Road Name
- Weather Condition

The assessment identifies missing timestamps, duplicate sensor readings, negative vehicle counts, unrealistic speeds, missing weather information, inconsistent road names, and sensor ID formatting errors.

3. Data Quality Issues

Issue	Example	Type
Missing timestamp	NULL	Missing
Duplicate sensor reading	S106	Duplicate
Negative vehicle count	-50	Invalid
Unrealistic speed	350 km/h	Outlier
Missing weather	NULL	Missing
Road variation	Anna Salai/anna salai	Inconsistency
Sensor ID error	S10A	Formatting

4. Cleaning Method

1. Duplicate sensor readings were removed.

2. Sensor ID formatting was standardized.
3. Timestamp values were converted into datetime format.
4. Negative vehicle counts were marked invalid.
5. Speeds above the specified threshold were treated as invalid.
6. Road names were standardized.
7. Missing weather values were replaced with Unknown.
8. Average speed and vehicle count were recalculated after cleaning.

5. Code Explanation

The code uses Pandas to process the time-series sensor data.

Duplicate readings are removed using `drop_duplicates()`.

Timestamps are converted using `pd.to_datetime()`.

Negative vehicle counts are treated as invalid.

The abnormal speed value of 350 km/h is identified as a faulty sensor reading.

Road names are standardized so that different representations refer to the same road.

The cleaned dataset is then used to calculate average speed and vehicle counts.

6. Output

```
Rows before cleaning: 10

===== DATA QUALITY CHECK =====
Missing Values:
SensorID      0
Timestamp     1
VehicleCount  0
AverageSpeed  0
TrafficSignalStatus 0
RoadName      0
WeatherCondition 2
dtype: int64

Duplicate Rows:
1

Average Speed Before Cleaning:
78.7 km/h

===== CLEANED DATASET =====
  SensorID  Timestamp  VehicleCount  AverageSpeed  TrafficSignalStatus  RoadName  WeatherCondition
0    S101  2025-01-01 08:00:00      120.0         45.0             Green    Anna Salai         Sunny
1    S102  2025-01-01 08:05:00      135.0         52.0             Red    Anna Salai         Sunny
2    S103  2025-01-01 08:10:00       NaN         48.0             Green    Anna Salai         Cloudy
3    S104  2025-01-01 08:15:00      140.0         NaN             Yellow   GST Road           Rainy
4    S105                NaT       110.0         42.0             Green   GST Road           Sunny
5    S106  2025-01-01 08:25:00      130.0         46.0             Red   Old Mahabalipuram Road      Unknown
7    S107  2025-01-01 08:30:00      150.0         55.0             Green   Old Mahabalipuram Road      Sunny
8    S108  2025-01-01 08:35:00      145.0         53.0             Green   Old Mahabalipuram Road      Cloudy
9    S109  2025-01-01 08:40:00      138.0         50.0             Green   GST Road           Rainy

Rows after cleaning: 9

===== TRAFFIC ANALYSIS =====
Average Speed Before Cleaning: 78.7 km/h
Average Speed After Cleaning: 48.88 km/h

Average Vehicle Count: 133.5

Average Vehicle Count by Road:
      VehicleCount
RoadName
Old Mahabalipuram Road  141.666667
GST Road                129.333333
Anna Salai              127.500000

dtype: float64

Cleaned dataset saved as: Q4_Traffic_cleaned.csv
```

7. Result

The traffic dataset was successfully cleaned by removing duplicate readings, identifying faulty sensor values, standardizing timestamps and road names, and handling missing weather information. The cleaned dataset provides a more reliable basis for congestion analysis.

QUESTION 5: GLOBAL SALES PERFORMANCE DATASET

1. Aim

To develop a data-cleaning framework for multinational sales data and standardize currencies, dates, products, salesperson names, missing values, ratings, and extreme sales values.

2. Dataset Description

The dataset contains:

- Region
- Country
- Salesperson
- Product
- Sales Amount
- Currency
- Sales Date
- Customer Rating

The assessment identifies multiple currencies, missing sales amounts, duplicate entries, invalid customer ratings, different date formats, multilingual product names, salesperson spelling variations, and extreme sales outliers.

3. Data Quality Issues

Issue	Example	Type
Multiple currencies	INR, USD, EUR, GBP	Standardization
Missing sales amount	NULL	Missing
Duplicate entry	Anna M	Duplicate
Invalid rating	6, 0	Invalid
Date variation	10-01-2025 / 01-15-2025	Formatting
Product variation	Laptop / Ordinateur Portable	Inconsistency
Salesperson variation	Raj Kumar / Rajkumar	Inconsistency
Extreme sales	9999999	Outlier

4. Cleaning Method

1. Duplicate records were removed.
2. Salesperson names were standardized.
3. Product names were translated/mapped to a common name.

4. Regions were standardized.
5. Sales dates were converted into a common datetime format.
6. Missing sales amounts were excluded from sales aggregation.
7. Customer ratings outside 1–5 were converted to missing values.
8. Sales amounts were converted to INR for comparison.
9. The extreme sales value was treated as an outlier for the demonstration.
10. Regional sales were calculated after cleaning.

5. Code Explanation

The code first reads the global sales CSV.

Different salesperson spellings are standardized using `replace()`.

Ordinateur Portable is mapped to Laptop.

Sales dates are converted using `pd.to_datetime()`.

Customer ratings are validated so that only values from 1 to 5 are considered valid.

Currency conversion is performed to provide a common reporting currency.

The regional sales values are grouped and sorted to identify the highest-performing region.

6. Output

```
===== SALES BEFORE CLEANING =====  
SalesAmountINR  
Region  
Asia      10139999.0  
Europe    333000.0  
North America 124500.0  
UK        115500.0  
dtype: float64
```

===== CLEANED DATASET =====									
	Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating	SalesAm
0	Asia	India	Raj Kumar	Laptop	50000.0	INR	2025-10-01	4.0	
1	Asia	India	Raj Kumar	Laptop	55000.0	INR	2025-10-01	5.0	
2	Europe	Germany	Anna M	Laptop	1200.0	EUR	2025-12-01	4.0	
4	North America	USA	John D	Laptop	1500.0	USD	2025-01-15	NaN	
6	Europe	France	Pierre L	Laptop	1300.0	EUR	2025-01-20	4.0	
8	Europe	UK	David T	Laptop	1100.0	GBP	2025-01-28	2.0	
9	Asia	India	Priya S	Smartphone	35000.0	INR	2025-01-30	NaN	
Rows after cleaning: 7									

7. Result

The multinational sales dataset was successfully cleaned by standardizing currencies, products, dates, regions, salesperson names, and customer ratings. Invalid and extreme values were handled, allowing regional sales performance to be compared more reliably.

OVERALL CONCLUSION

The five datasets demonstrated common real-world data-quality problems including missing values, duplicate records, invalid values, outliers, inconsistent formats, incorrect dates, and inconsistent categorical values.

The data-cleaning process improved the reliability of the datasets by:

- Removing duplicate records.
- Handling missing values.
- Detecting invalid values.

- Treating extreme values appropriately.
- Standardizing categorical data.
- Converting dates into consistent formats.
- Converting monetary values into a common currency for comparison.
- Generating meaningful analytical results.

The exercise demonstrates that data cleaning is an essential preprocessing step before performing data analysis or making business decisions. Incorrect or inconsistent data can produce misleading revenue calculations, hospital statistics, loan decisions, traffic analysis, and regional sales rankings.